B020314(020)

B. Tech. (Third Semester) Examination, Nov.-Dec. 2022

(New Scheme)

(Civil Engg. Branch)

PLANE SURVEYING

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note *: Attempt all questions. Part (a) is compulsory from each question and carries 4 marks. Attempt any* ***two*** *parts from (b), (c) and (d) of each question which carry 8 marks each.*

Unit-I

1. (a) What are the fundamental lines of Dumpy level and their relationship? 4

(b) The error due to inclination of line of collimation is completely eliminated if back sight and foresight distances are equalized. Justify the statement. 8

(c) Write short notes on : (any **two**) 8

(i) Reciprocal Leveling

(ii) Fly Leveling

(iii) Profile Leveling

(d) The following refer to the reciprocal levels taken with one level :

Instrument	Staff reading on		Remarks
	A	B	
A	1·56	2·597	Distance between A & B = 1200 m
B	0·987	2·418	RL of A = 625·500 m

Is the instrument in adjustment? To what reading should the line of collimation be adjusted when the instrument is at B? Find RL of B. 8

Unit-II

2. (a) What do you mean by the term interpolation of contour? 4

(b) What are the characteristics of contours? Explain with neat sketch. 8

(c) Explain in detail the indirect method of locating of contours. 8

(d) Draw neat sketch and explain working of any **two** minor instruments : 8

(i) Abney level

(ii) Pantagraph

(iii) Ceylon Ghat Tracer

Unit-III

3. (a) Write the various sources of errors in Theodolite. 4

(b) Explain the method of reiteration for setting out horizontal angle. 8

(c) Explain the process of measurement of deflection angles. 8

(d) Write short notes on (any **two**) : 8

(i) Face right and face left

(ii) Swinging the telescope

(iii) Transiting the telescope

Unit-IV

4. (a) Write the various accessories used in plane table survey. 4

(b) Name the common methods of balancing the traverse. How is closing error in a traverse balanced? 8

(c) What is two points problem? Explain by neat sketch by applying of plane table work. 8

(d) What do you understand by the term orientation in plane tabling? 8

Unit-V

5. (a) What is the purpose of providing curve? 4

(b) Explain the process of setting out a simple circular curve using the method of offset from chord produced. 8

(c) What are the elements of compound curve? Explain with neat sketch. 8

(d) Derive the elements to set a reverse curve when the straight are non-parallel. 8